

1 Randomization and Derandomization

1.1 The indicator-variable proof template

For a maximization objective with nonnegative local rewards, write

$$\text{ALG} = \sum_i w_i I_i, \quad \mathbb{E}[\text{ALG}] = \sum_i w_i \Pr[I_i = 1].$$

Linearity of expectation does *not* require the indicators to be independent. Independence is needed only when a probability calculation multiplies probabilities of separate random choices. If every reward is collected with probability at least α and $W = \sum_i w_i \geq \text{OPT}$, then $\mathbb{E}[\text{ALG}] \geq \alpha \text{OPT}$. This guarantees the existence of a good outcome, not that every run is good. An expected approximation bound alone is not a high-probability bound.

1.2 Weighted MAX-CUT, MAX- k -CUT, and MAX-DICUT

Assume nonnegative edge weights and remove self-loops, which can never cross a cut. For undirected MAX- k -CUT, independently label every vertex uniformly from $\{1, \dots, k\}$, for $k \geq 2$. Count edges whose endpoint labels differ. For every edge $e = \{u, v\}$,

$$\Pr[I_e = 1] = 1 - \sum_{a=1}^k \Pr[u = a, v = a] = 1 - \frac{1}{k}.$$

Consequently

$$\mathbb{E}[\text{ALG}] = (1 - 1/k)W \geq (1 - 1/k) \text{OPT}.$$

The ordinary MAX-CUT case is $k = 2$, giving $\text{OPT}/2$; MAX-3-CUT gives $2\text{OPT}/3$. A run takes $O(n + m)$ time including evaluation.

For directed MAX-DICUT, select each vertex into S independently with probability $1/2$, and count only arcs from S to $V \setminus S$. Each arc (u, v) is counted with probability $1/4$, so $\mathbb{E}[\text{ALG}] = W/4 \geq \text{OPT}/4$. Do not count arcs in both directions: doing so changes the objective. LP rounding improves the retained fraction to $1/2$ (Section ??).

1.3 Conditional expectation: how to remove randomness

Let F be the expected final objective conditional on the decisions already fixed. If the next random variable has outcomes a with probabilities p_a , then

$$F = \sum_a p_a F_a, \quad F_a = \mathbb{E}[\text{ALG} \mid \text{fixed decisions}, X = a].$$

At least one positive-probability branch has $F_a \geq F$. Fix such a branch and repeat. The invariant is that the conditional expectation never decreases; after the last decision it equals the actual objective. Thus the deterministic output is at least the initial expectation. To claim a polynomial-time algorithm, show how to compute each F_a in polynomial time.

MAX- k -CUT: computable conditional contributions

Under a partial labelling, an edge with both endpoints fixed contributes its full weight if their labels differ and zero otherwise. An edge with at least one endpoint unfixed contributes $(1 - 1/k)w_e$. When assigning a vertex v , only edges to already fixed neighbors have label-dependent contributions. Choose a label a maximizing

$$\sum_{u: \{u, v\} \in E, u \text{ fixed, label}(u) \neq a} w_{uv},$$

equivalently minimizing the weight of fixed neighbors with label a . Process vertices in any order. This gives a deterministic $(1 - 1/k)$ retained-value approximation in $O(m + nk)$ time by accumulating neighbor weights in k buckets at each vertex.

Worked MAX-CUT example. For a unit-weight triangle the initial expectation is $3/2$. Fix the first vertex to side 0; the expectation remains $3/2$. Putting the second on side 1 makes their edge contribute 1 and the two unresolved edges contribute $1/2$ each, giving conditional expectation 2. Either choice for the third vertex then yields cut value 2.

MAX-SAT: computable conditional contributions

First remove duplicate literals, discard empty clauses (always false), and separate tautologies (always true). Thus every remaining clause contains distinct variables and no variable together with its negation. Suppose an unfixed variable x_j is independently true with probability p_j . For each clause C_i , its conditional expected contribution is

$$\begin{cases} w_i, & \text{a fixed literal is already true,} \\ 0, & \text{all literals are fixed and false,} \\ w_i \left(1 - \prod_{\ell \text{ unfixed in } C_i} q_\ell \right), & \text{otherwise,} \end{cases}$$

where $q_{x_j} = 1 - p_j$ and $q_{\neg x_j} = p_j$ are literal-failure probabilities. For fair variables and r unfixed literals, the last line is $w_i(1 - 2^{-r})$. Evaluate the total for setting the next variable to 0 and to 1, and keep the larger value. Scanning all L literal occurrences for each of n variables gives $O(nL)$ arithmetic operations. This works for fair, biased, and LP-derived independent probabilities.

Worked example. For unit-weight clauses (x) and $(\neg x \vee y)$ with fair variables, $F = 1/2 + 3/4 = 5/4$. Setting $x = 1$ gives $F_1 = 1 + 1/2 = 3/2$; setting $x = 0$ gives $F_0 = 0 + 1 = 1$. Choose $x = 1$, then $y = 1$, obtaining value 2.

For a directed cut with arbitrary independent membership probabilities, an arc's conditional contribution is $w_{uv}p'_u(1 - p'_v)$, where p'_v is 0 or 1 for a fixed vertex and its original probability otherwise. This also makes both MAX-DICUT rounding schemes below derandomizable.

1.4 Weighted MAX-SAT by fair and biased assignment

Use the clause preprocessing above; handle tautological clauses as a constant reward added to every solution. All weights are nonnegative. For a remaining clause C_i of length $\ell_i \geq 1$, fair independent assignment gives

$$\Pr[C_i \text{ satisfied}] = 1 - 2^{-\ell_i}, \quad \mathbb{E}[\text{ALG}] = \sum_i w_i(1 - 2^{-\ell_i}) \geq W/2 \geq \text{OPT}/2.$$

The satisfaction probability *increases* with clause length. If every clause has at least k literals, the guarantee strengthens to $(1 - 2^{-k}) \text{OPT}$. An assignment and its value can be computed in $O(n + L)$ time, where L is the number of literal occurrences.

Biased assignment and the unit-clause issue

Merge repeated unit clauses by adding their weights. For each variable, rename it consistently throughout the formula if needed so that the total positive unit-clause weight is at least the negative one. Let N be the sum of the negative unit-clause weights after this renaming. Every assignment loses at least the smaller weight in each opposing pair, so $\text{OPT} \leq W - N$.

Set every renamed variable independently true with probability $p \in [1/2, 1]$. A clause with a positive and b negative literals is satisfied with probability

$$1 - (1 - p)^a p^b \geq 1 - p^{a+b}.$$

In a lower bound, ignore the negative unit clauses. Every remaining unit clause is satisfied with probability p , and every nonunit clause with probability at least $1 - p^2$. Therefore

$$\mathbb{E}[\text{ALG}] \geq \min\{p, 1 - p^2\}(W - N) \geq \min\{p, 1 - p^2\} \text{OPT}.$$

The maximum occurs when $p = 1 - p^2$, giving

$$p = \frac{\sqrt{5} - 1}{2}, \quad \mathbb{E}[\text{ALG}] \geq \frac{\sqrt{5} - 1}{2} \text{OPT}.$$

If no negative unit clauses remain, this simplifies to $N = 0$. Without the renaming/upper-bound argument, assuming all unit clauses are positive is not justified. Conditional expectation gives deterministic versions of these guarantees.

2 Linear Programming and Rounding

2.1 Modelling, relaxation, and the direction of the bound

An ILP uses integer variables; a 0–1 formulation uses binary decision variables. A relaxation replaces $x \in \{0, 1\}$ by $0 \leq x \leq 1$ and keeps the linear constraints. Its feasible set contains all integral feasible points. Write LP^* for the *optimal* relaxed objective:

	relaxation bound	target rounding proof
minimization	$LP^* \leq \text{OPT}$	$\text{ALG} \leq \rho LP^* \leq \rho \text{OPT}$
maximization	$LP^* \geq \text{OPT}$	$\mathbb{E}[\text{ALG}] \geq \alpha LP^* \geq \alpha \text{OPT}$

An arbitrary feasible fractional objective is not necessarily a valid bound on OPT in the displayed direction. LP is solvable in polynomial time in its encoding length; general ILP is NP-hard. After rounding, prove *feasibility* separately from the objective bound.

Proving a model is exact. Show both (i) each original solution gives a feasible integral point of the same objective and (ii) an integral point decodes into a solution of at least its objective for maximization (at most for minimization). These establish equality of optimal values. An auxiliary reward variable often needs only upper bounds: maximizing a nonnegative weighted objective pushes it up when possible. Such constraints need not force its intended Boolean meaning at *every feasible point*. If its coefficient is zero, it need not be pushed up even at an optimum; the decoded solution and optimal value remain correct. Add equivalence constraints if the variable’s exact meaning is required independently of the objective.

2.2 Weighted vertex cover: threshold rounding

For nonnegative vertex costs c_v , let x_v indicate selection. The ILP is

$$\min \sum_{v \in V} c_v x_v \quad \text{subject to} \quad x_u + x_v \geq 1 \quad (\{u, v\} \in E), \quad x_v \in \{0, 1\}.$$

Relax to $[0, 1]$, obtain an optimum x^* , and return $C = \{v : x_v^* \geq 1/2\}$. At least one endpoint of every edge reaches $1/2$, so C is feasible. Also

$$c(C) = \sum_{v \in C} c_v \leq \sum_{v \in C} 2c_v x_v^* \leq 2LP^* \leq 2\text{OPT}.$$

The algorithm is polynomial time (LP plus a linear scan). On the unweighted complete graph K_n , $\text{OPT} = n - 1$ and $LP^* = n/2$: the all- $1/2$ point is feasible, and summing all edge constraints gives $(n - 1) \sum_v x_v \geq n(n - 1)/2$. Hence the worst-case integrality gap is at least $2 - 2/n$, approaching 2. This is a family of gap examples, not a claim that every n -vertex graph has this gap.

2.3 Set cover: bounded frequency

Let U be a universe, $S_1, \dots, S_m$ candidate sets with $\bigcup_j S_j = U$, and costs $c_j \geq 0$. One binary variable per set suffices:

$$\min \sum_{j=1}^m c_j x_j \quad \text{subject to} \quad \sum_{j: e \in S_j} x_j \geq 1 \quad (e \in U), \quad x_j \in \{0, 1\}.$$

There is no need for an extra variable indicating coverage when every element must be covered. If each element belongs to at most k sets, relax and select every set with $x_j^* \geq 1/k$. Each covering constraint has at most k terms, so at least one reaches the threshold. The cost satisfies

$$\text{ALG} \leq k \sum_j c_j x_j^* = kLP^* \leq k\text{OPT}.$$

This generalizes vertex-cover rounding, where an edge has two endpoint choices. The hypothesis is bounded *frequency*, not bounded set size. For example, with one element and two identical cost-1 singleton sets, $(1/2, 1/2)$ is LP-optimal. Each set has size 1, but threshold 1 selects nothing. This disproves the corresponding threshold proof for bounded set size.

2.4 Weighted MAX-SAT: LP rounding

For the preprocessed formula of Section ??, use x_j for a variable's truth value and y_i for the reward of clause C_i :

$$\begin{aligned} \max \quad & \sum_i w_i y_i \\ \text{subject to} \quad & y_i \leq \sum_{x_j \in C_i} x_j + \sum_{\neg x_j \in C_i} (1 - x_j) \quad \text{for every } i, \\ & x_j, y_i \in \{0, 1\}. \end{aligned}$$

A false clause forces $y_i = 0$; a true clause permits $y_i = 1$. Nonnegative weights make the ILP optimal value exactly the maximum satisfied weight. Relax all variables to $[0, 1]$ and write $LP^* = \sum_i w_i y_i^* \geq \text{OPT}$. Independently set x_j true with probability x_j^* .

One-clause bound: product, AM–GM, then concavity

For a clause of length $\ell \geq 1$, let $a_1, \dots, a_\ell$ be its literal-success probabilities: x_j^* for a positive literal and $1 - x_j^*$ for a negative one. Its variables are distinct, and $\sum_h a_h \geq y_i^*$. Hence

$$\begin{aligned} \Pr[C_i \text{ satisfied}] &= 1 - \prod_{h=1}^{\ell} (1 - a_h) \\ &\geq 1 - \left(1 - \frac{\sum_h a_h}{\ell}\right)^\ell \geq 1 - \left(1 - \frac{y_i^*}{\ell}\right)^\ell. \end{aligned}$$

The exponent is ℓ , *not* $1/\ell$. Let $g_\ell(y) = 1 - (1 - y/\ell)^\ell$. For $\ell = 1$ it is linear; for $\ell \geq 2$,

$$g_\ell''(y) = -\frac{\ell-1}{\ell} (1 - y/\ell)^{\ell-2} \leq 0.$$

Since $g_\ell(0) = 0$, concavity on $[0, 1]$ gives $g_\ell(y) \geq y g_\ell(1)$. Also $(1 - 1/\ell)^\ell \leq e^{-1}$. Thus, with $b_\ell = 1 - (1 - 1/\ell)^\ell$,

$$\Pr[C_i \text{ satisfied}] \geq b_\ell y_i^* \geq (1 - 1/e) y_i^*, \quad \mathbb{E}[\text{ALG}] \geq (1 - 1/e) LP^* \geq (1 - 1/e) \text{OPT}.$$

The LP can be solved in polynomial time; sampling and evaluating the result takes $O(n + L)$ time. The conditional-product calculation in Section ?? derandomizes the algorithm.

Combine fair assignment and LP rounding: 3/4

For a clause of length ℓ , fair assignment satisfies it with probability $a_\ell = 1 - 2^{-\ell} \geq a_\ell y_i^*$, since $0 \leq y_i^* \leq 1$. LP rounding gives probability at least $b_\ell y_i^*$. The useful comparison is clause by clause:

ℓ	a_ℓ	b_ℓ	$a_\ell + b_\ell$
1	1/2	1	3/2
2	3/4	3/4	3/2
$\ell \geq 3$	$\geq 7/8$	$\geq 1 - 1/e > 5/8$	$> 3/2$

Run both algorithms and return the better assignment. If their values are A, B , then pointwise $\max\{A, B\} \geq (A + B)/2$, so

$$\mathbb{E}[\max\{A, B\}] \geq \frac{1}{2} \sum_i w_i (a_{\ell_i} + b_{\ell_i}) y_i^* \geq \frac{3}{4} LP^* \geq \frac{3}{4} \text{OPT}.$$

The two outputs need not be independent. Averaging the global constants $1/2$ and $1 - 1/e$ would not prove $3/4$; the complementary clause-length bounds are essential. Alternatively derandomize each algorithm first and take the better: the same $3/4$ guarantee then holds deterministically.

2.5 MAX-DICUT: two 1/2 rounding proofs

Let $G = (V, A)$ be a directed graph with nonnegative arc weights and no self-loops. Put $x_i = 1$ when vertex i is on the source side. For each arc, use a reward variable z_{ij} . The ILP and its relaxation are

$$\begin{aligned} \max \quad & \sum_{(i,j) \in A} w_{ij} z_{ij} \\ \text{subject to} \quad & z_{ij} \leq x_i, \quad z_{ij} \leq 1 - x_j, \\ & x_i, z_{ij} \in \{0, 1\} \quad (\text{replace by } [0, 1] \text{ for the LP}). \end{aligned}$$

The upper bounds allow a positive reward only when $x_i = 1, x_j = 0$. For any fixed integral cut, setting $z_{ij} = 1$ on its outgoing arcs attains the cut weight, so the ILP is exact. If exact equality $z_{ij} = x_i(1 - x_j)$ is required at every integral feasible point, add $z_{ij} \geq x_i - x_j$; it is unnecessary for this nonnegative maximization objective. Below, x, z denote an optimal solution of the displayed relaxation and $L = LP^* \geq \text{OPT}$.

Shifted probabilities: a per-arc comparison

Independently select vertex i with probability $p_i = 1/4 + x_i/2$. For each arc,

$$\begin{aligned} \Pr[i \in S, j \notin S] &= (1/4 + x_i/2)(1/4 + (1 - x_j)/2) \\ &\geq (1/4 + z_{ij}/2)^2 \\ &= z_{ij}/2 + (2z_{ij} - 1)^2/16 \geq z_{ij}/2. \end{aligned}$$

Summing gives $\mathbb{E}[\text{ALG}] \geq L/2 \geq \text{OPT}/2$. This per-arc inequality holds for any feasible LP point, though optimality (or another appropriate lower bound on its value) is needed to compare with OPT.

Direct probabilities: a global comparison

Rounding instead with $p_i = x_i$ also gives a 1/2 guarantee for an *optimal* solution of this LP. Let $W = \sum_{(i,j) \in A} w_{ij}$. If $W = 0$, any cut is optimal. Otherwise,

$$\Pr[i \in S, j \notin S] = x_i(1 - x_j) \geq z_{ij}^2.$$

Weighted Cauchy–Schwarz gives

$$L^2 = \left(\sum_{ij} \sqrt{w_{ij}} z_{ij} \sqrt{w_{ij}} \right)^2 \leq \left(\sum_{ij} w_{ij} z_{ij}^2 \right) W.$$

The feasible point $x_i = 1/2, z_{ij} = 1/2$ has value $W/2$, so optimality implies $L \geq W/2$. Consequently

$$\mathbb{E}[\text{ALG}] \geq \sum_{ij} w_{ij} z_{ij}^2 \geq L^2/W \geq L/2 \geq \text{OPT}/2.$$

Direct rounding lacks the shifted scheme's per-arc bound $z_{ij}/2$, but its global analysis recovers the same factor. Both algorithms are polynomial time and can be derandomized using conditional expectation.

Tuning an affine probability safely

Let $p_i = a + bx_i$, with $a, b \geq 0$ and $a + b \leq 1$, so all probabilities and all factors below are nonnegative. Write $c = 1 - a - b \geq 0$. Then

$$p_i(1 - p_j) \geq (a + bz_{ij})(c + bz_{ij}) = ac + b(1 - b)z_{ij} + b^2 z_{ij}^2.$$

For fixed b , $a + c = 1 - b$ and ac is largest when $a = c = (1 - b)/2$. For this symmetric choice,

$$(a + bz)^2 = 4abz + (a - bz)^2 \geq 4abz, \quad \alpha = 4ab = 2b(1 - b) = \frac{1}{2} - 2(b - \frac{1}{2})^2 \leq \frac{1}{2}.$$

Maximize the retained fraction α : choose $b = 1/2, a = 1/4$, obtaining 1/2. This optimizes this particular square-based lower bound. It does not by itself prove that every other analysis of affine rounding is incapable of a stronger guarantee.

2.6 Randomized set cover: repeat until feasible

Use the set-cover LP in Section ??, without a frequency bound. Let $n = |U| \geq 1$, m be the number of candidate sets, and $r = \lceil \ln(2n) \rceil$. (For $U = \emptyset$, return the empty cover.) Solve the LP once. A *trial* performs r independent rounds; in each round select each set S_j independently with probability x_j^* , then take the union of all selected sets. Check coverage. Discard a failed trial and start a fresh independent trial until a cover is found.

Success probability. For any element e , the probability it remains uncovered in one round is

$$\prod_{j:e \in S_j} (1 - x_j^*) \leq \exp \left(- \sum_{j:e \in S_j} x_j^* \right) \leq e^{-1}.$$

After r independent rounds its failure probability is at most $e^{-r} \leq 1/(2n)$. By the union bound, a trial succeeds with probability $q \geq 1/2$. Element-failure events need not be independent.

Cost of a trial and cost of the returned trial. A set is included in the union with probability $1 - (1 - x_j^*)^r \leq r x_j^*$. Thus its nonnegative total cost C satisfies $\mathbb{E}[C] \leq rLP^*$. However, successful trials can cost more than typical trials. If A is the success event, the returned trial has the distribution of one trial conditioned on A , and

$$\mathbb{E}[C_{\text{returned}}] = \mathbb{E}[C \mid A] = \frac{\mathbb{E}[C \mathbf{1}_A]}{q} \leq \frac{\mathbb{E}[C]}{q} \leq 2rLP^* \leq 2r \text{OPT}.$$

Fresh independent trials have a geometric stopping time T with $\Pr[T > t] = (1 - q)^t$ and $\mathbb{E}[T] = 1/q \leq 2$. Hence the algorithm terminates almost surely, always returns a checked feasible cover, and takes expected polynomial time. One trial uses $O(rm)$ random decisions and $O(n + \sum_j |S_j|)$ time to check coverage after marking selected sets. The cost guarantee is in expectation; it is not a bound on every returned cover. The factor $1/q$ cannot be dropped without an additional argument, because cost and success may be correlated.